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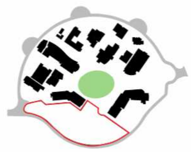
CONCEPT DESIGN IMAGERY

LIVING LOOP

The Mission College STEM building is a continuous, connected system, inspired by the geometry of the Möbius strip. It enables seamless transitions between spaces and disciplines, blurring the line between indoor and outdoor environments to support a dynamic, evolving learning experience.

Through a looped circulation strategy, students flow between labs, classrooms, and outdoor learning environments—spaces that extend into the landscape through canopies, courtyards, and planted zones. The architecture integrates natural daylight, passive systems, and biophilic design, leveraging the unique climate and ecological identity of Santa Clara to create a resilient, regenerative, and future-ready learning environment.



CONTEXT

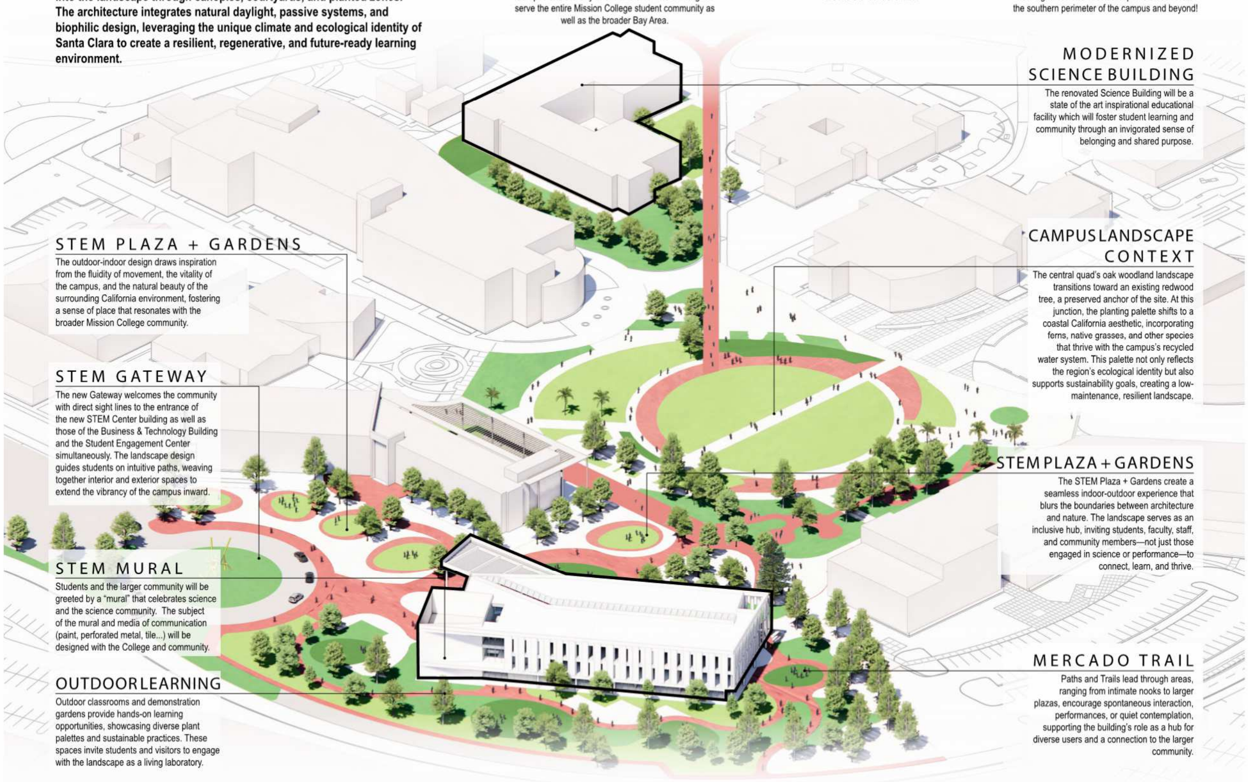
The STEM Building will be an academic and social catalyst in the context of the beautiful Mission College Campus and the City of Santa Clara. The building will serve the entire Mission College student community as well as the broader Bay Area.

CONNECT

The STEM Building will complete - and visually connect - the circle of buildings that are centered around the Central Plaza.

ACTIVATE

The orientation and primary entrances of the STEM Building both reinforce as well as create new sightlines and circulation paths that activate the southern perimeter of the campus and beyond!



MODERNIZED SCIENCE BUILDING

The renovated Science Building will be a state of the art inspirational educational facility which will foster student learning and community through an invigorated sense of belonging and shared purpose.

STEM PLAZA + GARDENS

The outdoor-indoor design draws inspiration from the fluidity of movement, the vitality of the campus, and the natural beauty of the surrounding California environment, fostering a sense of place that resonates with the broader Mission College community.

CAMPUS LANDSCAPE CONTEXT

The central quad's oak woodland landscape transitions toward an existing redwood tree, a preserved anchor of the site. At this junction, the planting palette shifts to a coastal California aesthetic, incorporating ferns, native grasses, and other species that thrive with the campus's recycled water system. This palette not only reflects the region's ecological identity but also supports sustainability goals, creating a low-maintenance, resilient landscape.

STEM GATEWAY

The new Gateway welcomes the community with direct sight lines to the entrance of the new STEM Center building as well as those of the Business & Technology Building and the Student Engagement Center simultaneously. The landscape design guides students on intuitive paths, weaving together interior and exterior spaces to extend the vibrancy of the campus inward.

STEM PLAZA + GARDENS

The STEM Plaza + Gardens create a seamless indoor-outdoor experience that blurs the boundaries between architecture and nature. The landscape serves as an inclusive hub, inviting students, faculty, staff, and community members—not just those engaged in science or performance—to connect, learn, and thrive.

STEM MURAL

Students and the larger community will be greeted by a "mural" that celebrates science and the science community. The subject of the mural and media of communication (paint, perforated metal, tile...) will be designed with the College and community.

MERCADO TRAIL

Paths and Trails lead through areas, ranging from intimate nooks to larger plazas, encourage spontaneous interaction, performances, or quiet contemplation, supporting the building's role as a hub for diverse users and a connection to the larger community.

OUTDOOR LEARNING

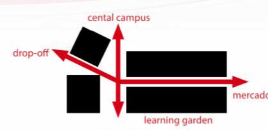
Outdoor classrooms and demonstration gardens provide hands-on learning opportunities, showcasing diverse plant palettes and sustainable practices. These spaces invite students and visitors to engage with the landscape as a living laboratory.

connected experience.

Develop outdoor spaces to enhance comfort and connection with nature.

CONTEXT

The STEM building is organized to maximize access and visibility to the STEM Ctr and Conference + Performance spaces. And, simultaneously optimize the flexibility and efficiency of the instructional spaces.



CONNECT

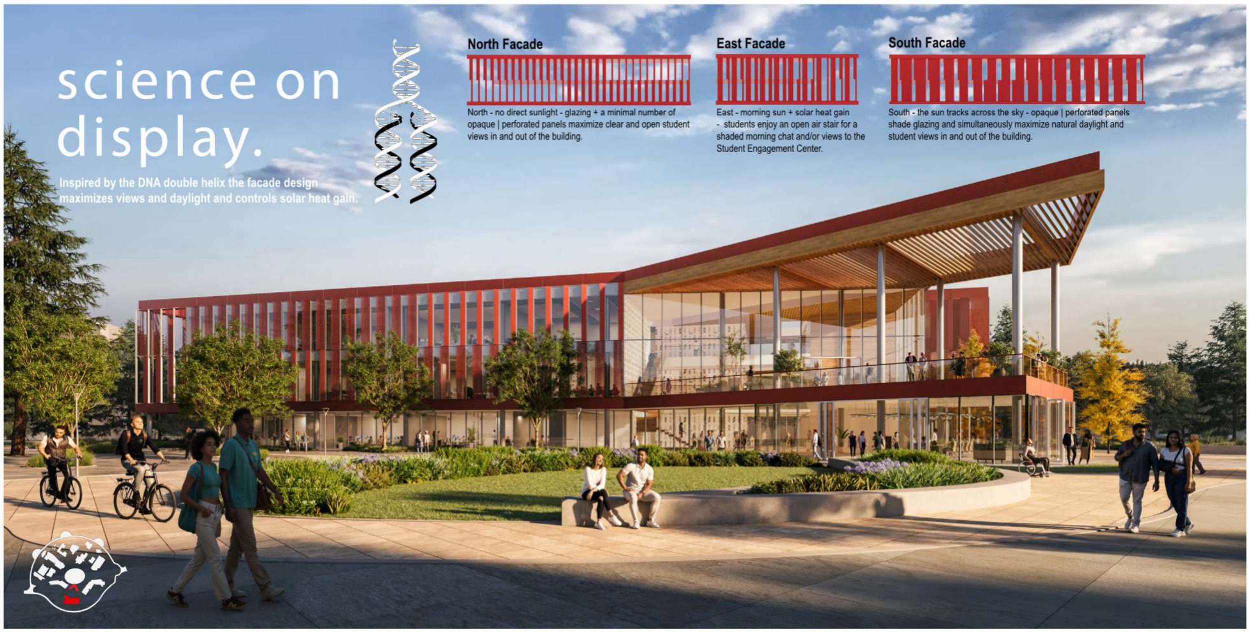
The STEM Building is shaped to fit into the larger context of the campus as a visible and accessible beacon from multiple vantage points.

ACTIVATE

The hub or central BRAIN of the STEM Building create active pathways which promote student interaction, connection and collaboration.

science on display.

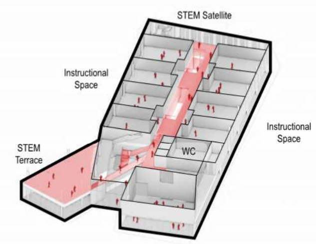
Inspired by the DNA double helix the facade design maximizes views and daylight and controls solar heat gain.



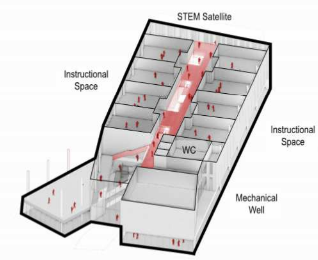
COMMUNITY SERVING.

Develop campus facilities to enhance collaborations, partnerships, and events.

L1



L2



L3

THE SOLAR ACCELERATOR

The sun tracks the Living Loop filtering daylight through all levels of the STEM building. Exposure to natural light will improve student's moods, reduce stress and create conducive learning environments.

SHADE canopy

The Shade Canopy is an open invitation to students from the entire campus to visit the STEM building as a social and academic destination. The welcoming trellis structure is a familiar campus architectural element, filtering the sun and offering protection from wind and rain.

conference + performance

Another highly flexible space, the conference and performance space includes a Servery, Storage and an AV room. The space which can also be used by the enterprise community will be for all-campus student used. Acoustics, lighting, equipment and furniture layouts will be designed to accommodate diverse needs such as academic lectures, performances, and "enterprise" conferences. Students will use a "balcony" at the second level as a peek through mezzanine to view activity in the space.

stem terrace

The STEM Terrace is easy to see and access from all points on campus. Students intuitively flow into the STEM building and on to the terrace as a natural outdoor extension of the STEM Center and Conference + Performance Space on the ground level. On the terrace students will have access to shelter, WIFI and each other! And, during campus events the Terrace will serve as a viewing platform back to the Central Plaza.

stem CEnter

The STEM Center space is a technically rich environment accessible to the entire campus community as a place for study and a resource for academic support. During the day, folding doors open into a learning garden and outdoor classroom. At night, the Center is a well-lit beacon, providing a safe and secure place for all students to learn and succeed.

the brain

"The Brain" is a highly flexible space which will be used as a lobby, a Pre-function area for the Conference + Performance Space, a breakout space for the STEM Center and, importantly, a comfortable, safe and inclusive space for students to rest and hangout between classes.

DYNAMIC INTERIOR ORGANIZATION.

Support interdisciplinary collaboration.

the living loop

An intuitive path draws students through the building, guided by natural light above. The loop, accessible to all, is not only a means to a destination but offers different experiences along the way. The Science Wall teaches and delights. And STEM innovation nooks embedded in the Science Wall are student collaboration areas.

innovation nooks

Embedded in the Science Wall, students literally inhabit the science they are learning about with immediate opportunities to hangout with other students to talk and share their ideas and knowledge.

science wall

Washed by natural light, the science wall is a dynamic collection of facts, figures, symbols, images and discovery. Art and Science intersect in this student curated Science Wall!

stem satellite

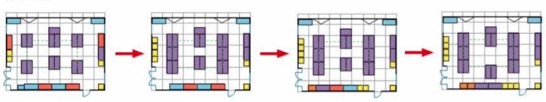
Part of the orbit of the Living Loop, the Satellite STEM area is a casual, comfortable space where students can study, have a quiet conversation, or even take a nap. From the STEM Satellite, students can see their peers on the deck of the Student Engagement Center across the way and open-air stairs descend to connect to the Mercado trail.

Flexible multi-use space

Encourage flexibility and adaptability by limiting single use spaces.

Lab layouts

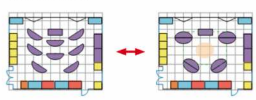
Containment



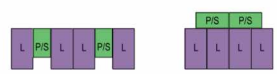
High | Medium | Low | None

Lab Benches

Student Collaboration



Lab prep/store



Segregated | Combined

maximize value.

Account for cost of ownership.

Living Loop

They step through the gateway, unsure where to start,
A question held closely, a spark in the heart.
The building responds—not with silence or stone,
But with rhythm and light, and a path of their own.
The loop is alive—its structure, its skin,
Draws the outside around them, and pulls them within.
Where gardens take root in the lessons they learn,
And leaves trace the cycles of how students turn.
Through biology's breath, through chemistry's flame,
Through soil and stem, they're never the same.
In learning that's lived—in motion, in place—
They test what they know, and find purpose in pace.
Canopies stretch where the classrooms unfold,
Letting sunlight and science and wonder take hold.
No hallway is simple, no corridor still—
Each one invites motion, intention, and will.
They move not in lines, but in layered refrain—
A loop that repeats but never the same.
The building expands with the mind at its core,
Each step through its form unlocking the door.
From roots in the ground to reactions in air,
To asking new questions, to daring to care—
They change through the loop, they rise into view,
Shaped by a space that believes in them too.